一、填空题

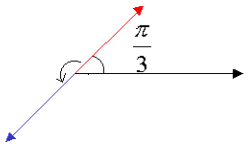
1．

$$v = v_m \cos(\omega t + \varphi + \frac{\pi}{2}) \qquad \frac{1}{2}v_m = v_m \cos(\varphi + \frac{\pi}{2})$$

$$\frac{\pi}{2} + \varphi = \begin{cases} \frac{\pi}{3} \\ -\frac{\pi}{3} \end{cases} \qquad \varphi = \begin{cases} -\frac{\pi}{6} \\ -\frac{5\pi}{6} \end{cases} \qquad \varphi = -\frac{5\pi}{6} \; or \; \frac{7\pi}{6}$$

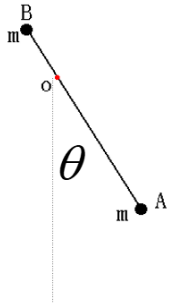
2.

$$\omega = 2\pi \qquad \omega\Delta t = \pi \qquad \Delta t = \frac{1}{2}$$



3.

$$T = 2\pi\sqrt{\frac{J}{mgh}}$$



$$J = Ml_1^2 + Ml_2^2 \qquad m = 2M \qquad h = \frac{l_1 - l_2}{2}$$

$$T = 2\pi\sqrt{\frac{l_1^2 + l_2^2}{(l_1 - l_2)g}}$$

$$-Mg(l_1 - l_2)\theta = J\alpha \qquad \frac{d^2\theta}{dt^2} + \frac{(l_1 - l_2)g}{l_1^2 + l_2^2}\theta = 0$$

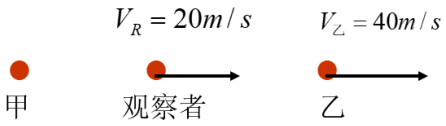
4.

$$t = \frac{T}{8}, \frac{3T}{8}$$

5.

$$\overline{P} = IS = wuS = \frac{wS\omega\lambda}{2\pi}$$

6.



$$\nu_{甲} = \frac{u - V_R}{u}\nu_s = 469.7HZ$$

$$\nu_{乙} = \frac{u + V_R}{u + V_{乙}}\nu_s = 472.97HZ$$

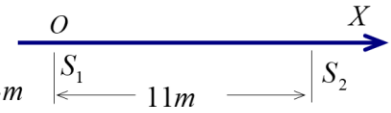
$$\Delta\nu = \nu_{乙} - \nu_{甲} = 3.3HZ$$

二、计算题

1 、 解：如图所示，取S_1点为坐标原点，S_1、S_2联线为X轴

$$\Delta\varphi = (\varphi_2 - \varphi_{1A}) - \frac{2\pi}{\lambda}(r_2 - r_1)$$

$$\lambda = \frac{u}{\nu} = 4m$$



(1) $x \le 0$ $r_1 = -x$ $r_2 = 11 - x$

$\Delta\varphi = -6\pi$ 没有因干涉而静止的点

(2) $0 \le x \le 11$ $r_1 = x$ $r_2 = 11 - x$

$$\Delta\varphi = -6\pi + \pi x = (2k+1)\pi$$

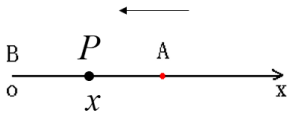
$x = 2k + 7$ $k = 2,1,0,-1,-2,-3$

(3) $x \ge 11$ $r_1 = x$ $r_2 = x - 11$

$\Delta\varphi = 5\pi$ 全部为干涉而静止的点

2、

(1) $y = 3.0\cos(4\pi t - 4\pi\frac{5-x}{20})$



$$y = 3.0\cos\left[4\pi(t + \frac{x}{20}) - \pi\right]$$

(2) $y = 3.0\cos 4\pi(t - \frac{x}{20})$

(3) $y = 6.0\cos(\frac{\pi x}{5} - \frac{\pi}{2})\cos(4\pi t - \frac{\pi}{2})$

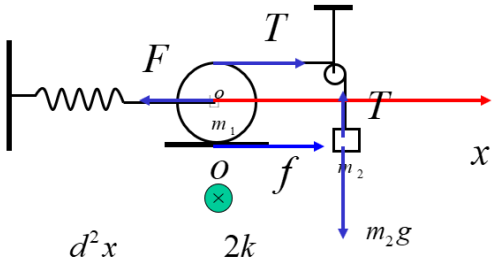
$$\frac{\pi x}{5} - \frac{\pi}{2} = k\pi$$

$x = 5k + 2.5$ $k = 0,1,2\cdots$

3、平衡时：$T = f = m_2 g$

$$T + f = kl_0 = 2m_2 g$$

$$l_0 = \frac{2m_2 g}{k}$$



C在任一位置：x

$$T + f - k(l_0 + x) = m_1 a_c$$

$$TR - fR = J_c \alpha = \frac{1}{2} m_1 R^2 \alpha$$

$$m_2 g - T = m_2 a$$

$$a_c = \alpha R$$

$$a = a_c + \alpha R = 2\alpha R$$

$$a_c = \frac{d^2 x}{dt^2} = -\frac{2k}{3m_1 + 8m_2} x$$

$$\frac{d^2 x}{dt^2} + \frac{2k}{3m_1 + 8m_2} x = 0$$

$$\omega = \sqrt{\frac{2k}{3m_1 + 8m_2}}$$